

# Operations Manual

How to run, administer, and maintain the PEDMAS platform

www.pedmas.com · 2026-08-21

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## 1. What you are operating

PEDMAS is an adaptive K–12 mathematics platform: 634 skills across Grades 1–12, 132 tap-through lessons, an adaptive placement test, daily practice with mastery tracking and spaced review, and a family subscription billed through Stripe.

|                          |                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Live site</b>         | <code>https://www.pedmas.com</code>                                                                      |
| <b>Local development</b> | <code>http://localhost:3080</code> (npm run dev in C:\Users\buruf\Documents\Pedmas)                      |
| <b>Code</b>              | <code>github.com/buruf/Pedmas</code> — every push to main deploys to production automatically via Vercel |
| <b>Database</b>          | Neon Postgres (production) / JSON files in data\ (local development)                                     |
| <b>Payments</b>          | Stripe — currently in SANDBOX (test) mode                                                                |
| <b>Email</b>             | Resend — key not yet configured; email features are dormant until it is                                  |

**Push to main = deploy to production. There is no staging environment. Run npm test and npm run build locally before pushing.**

## 2. Accounts and roles

- PARENT — created at signup by an adult. Owns child profiles, billing, and consent. Sees the parent dashboard and can delete any child profile or the whole account.
- STUDENT — a child profile inside a parent account. Children never have their own credentials or contact details.
- ADMIN — the platform owner (you). Sees everything at /admin plus all student profiles.

### The admin login

The admin account is created automatically the first time the app runs with no admin present. Its credentials come from two environment variables:

|                              |                                                                  |
|------------------------------|------------------------------------------------------------------|
| <b>PEDMAS_ADMIN_EMAIL</b>    | Admin email. Development fallback: <code>admin@pedmas.com</code> |
| <b>PEDMAS_ADMIN_PASSWORD</b> | Admin password. Development fallback: <code>pedmas-admin</code>  |

On your local machine the fallbacks work: log in at <http://localhost:3080/login> with `admin@pedmas.com` / `pedmas-admin`. In production the fallback has been verified NOT to work, which means your live admin uses the values stored in Vercel > your project > Settings > Environment Variables. If you have forgotten the password, look it up there.

**Changing the environment variable later does NOT change an existing admin account — the account is only seeded once. To actually change the admin password: use the password-reset email flow (requires Resend), or edit the account row in Neon directly.**

## 3. The admin console (/admin)

Log in as the admin, then open /admin. Non-admin accounts get “Admin only”. The console has five areas, top to bottom:

### 3.1 Overview cards

Accounts, Students, Placed students, Grades, Strands, Skills. Quick pulse of how much of the platform is in use.

### 3.2 Question generator preview

Pick any grade, skill and stage and see freshly generated questions exactly as students would, plus a health verdict (healthy / thin / very thin / fails validation) based on sampling: how many distinct questions the generator can produce and whether any fail validation. Use this when a parent reports a strange question — reproduce it here first.

### 3.3 Students table

Every student profile: grade, placement status, sessions completed, skills mastered, skills flagged struggling, and streak. Admin sees all students across all accounts.

### 3.4 Errors panel

- Every unhandled error — on the server or in a visitor's browser — is recorded automatically and grouped, with a count, the path, and when it last happened.
- “Fire a test error” throws a real server error on purpose. If it appears in the panel a moment later, monitoring is working end to end. Dismiss it afterwards.
- Dismiss removes a group once you have dealt with it. It reappears (count restarts) if the error happens again.
- When Resend is configured, a brand-new error also emails the admin address — at most one email every 6 hours.

### 3.5 Lesson effectiveness

For each lesson: first-try accuracy on that lesson's skills after children completed the lesson, versus a baseline (attempts before the lesson, plus children who never opened it), and the lift between them. Verdicts are withheld until each side has 25 attempts. Read it as a signal, not a controlled experiment — children choose whether to open lessons.

## 4. Everyday operations

### Opening and closing registration

Registration is CLOSED by default and the homepage shows an under-construction banner. To open it: in Vercel add `REGISTRATION_OPEN=true` (exactly the word `true`) and redeploy. Remove it (or set anything else) to close again. Existing accounts always keep working either way.

### Deploying

- Commit and push to main. Vercel builds and deploys automatically (about 2 minutes).
- Before pushing, always run locally: `npm test` (139 tests) and `npm run build`.
- Never run `npm run build` while the local dev server is running — it corrupts the dev server's cache and buttons silently stop working. If that happens: stop the dev server, delete the `.next` folder, start it again.

### Weekly parent email

A Vercel cron calls `/api/cron/weekly-progress` every Sunday 16:00 UTC. It emails each parent a summary of every child's week, honours the account-level opt-out and the one-click unsubscribe link, and sends at most one email per account per day even if the cron retries. It does nothing until the Resend key is set.

### Billing (Stripe)

- Pricing: \$11.99/month first child + \$5.99 each additional, up to 4 children, 7-day free trial.
- Currently in sandbox: use test card 4242 4242 4242 4242, any future expiry, any CVC.
- Parents manage cards, plan changes and cancellation themselves through the Stripe customer portal (Billing page in the app).
- Deleting an account automatically cancels its Stripe subscription immediately. If Stripe is unreachable at that moment, the deletion still happens and a loud log line tells you to cancel manually in the Stripe dashboard.
- Going live later requires: live Stripe keys + live webhook secret in Vercel, live price IDs, and the legal checklist below.

### Data and deletion promises

- The privacy policy promises immediate, real deletion. The app delivers it: parents can delete a child profile or the whole account from the Account page; both are immediate hard deletes.
- The database is Neon Postgres: per-entity tables (accounts, students, auth\_sessions, password\_reset\_tokens, rate\_limits, stripe\_events, error\_events). The old `pedmas_rows` table is a frozen pre-migration backup — never write to it and never re-import it (that would resurrect deleted data).
- Region (US or international wording/units) is detected from the family's location on first visit and stamped on the account. There is no UI to change it yet; if a family asks, edit their account row's region field in Neon (US or INTL).

## 5. Environment variables (Vercel)

|                                       |                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>DATABASE_URL</b>                   | Neon Postgres connection string. Without it, production cannot store anything.                                                |
| <b>PEDMAS_ADMIN_EMAIL / _PASSWORD</b> | Admin credentials, used once at first seeding.                                                                                |
| <b>REGISTRATION_OPEN</b>              | “true” opens signups. Anything else (or unset) keeps them closed.                                                             |
| <b>AUTH_COOKIE_DOMAIN</b>             | .pedmas.com — makes login work across pedmas.com and www.pedmas.com. Do not remove.                                           |
| <b>NEXT_PUBLIC_APP_URL</b>            | https://www.pedmas.com — used in emails and Stripe redirect URLs.                                                             |
| <b>STRIPE_SECRET_KEY</b>              | Stripe API key (currently sandbox sk_test_...).                                                                               |
| <b>STRIPE_WEBHOOK_SECRET</b>          | Signing secret for the Stripe webhook endpoint.                                                                               |
| <b>STRIPE_PRICE_FIRST_CHILD</b>       | Stripe price ID for the \$11.99 seat.                                                                                         |
| <b>STRIPE_PRICE_ADDITIONAL_CHILD</b>  | Stripe price ID for the \$5.99 seat.                                                                                          |
| <b>RESEND_API_KEY</b>                 | Resend key. NOT YET SET — password reset, weekly summaries, placement report emails and error alerts are dormant until it is. |
| <b>EMAIL_FROM</b>                     | From address for outgoing mail, e.g. PEDMAS <hello@pedmas.com>.                                                               |
| <b>CRON_SECRET</b>                    | Bearer token protecting the weekly cron and the health check. Both refuse to run without it.                                  |

## 6. When something breaks

### First steps

- The Errors panel at /admin — server and browser errors land there with counts and paths.
- Vercel dashboard > the project > Logs, for anything that crashed before the monitor could record it.
- The health check proves the store accepts writes: send GET https://www.pedmas.com/api/health with header Authorization: Bearer <CRON\_SECRET>.

### Rolling back a bad deploy

Two options: in the Vercel dashboard, promote the previous deployment (instant); or locally run git revert <bad-commit> and push, which redeploys the previous behaviour with history intact. Prefer revert — it keeps git and production telling the same story.

### Known local-development gotchas

- npm run build while the dev server runs breaks the dev server silently (dead buttons, hydration gone). Stop server, delete .next, restart.

- Never bulk-edit source files with PowerShell text commands — they corrupt em dashes and curly quotes. Use an editor.
- Local data lives in data\\*.json. Deleting those files resets your local accounts and students (production is untouched).

## 7. Before taking real money (legal checklist)

Tracked in detail in docs/LEGAL-RATIONALE.md in the repository. The blockers, in order:

- Fill the operator placeholders in src/lib/legal.ts: legal entity, postal address, monitored privacy email, governing jurisdiction. Counsel should advise on incorporating first.
- Accept the standard Data Processing Agreements with Stripe, Neon, Vercel and Resend (all self-serve).
- Decide launch geography (recommendation: US + Canada only) and state it in the terms.
- Have a lawyer review the Terms and Privacy Policy, then remove the draft banners.
- Write the short data-retention policy and information-security document the amended COPPA rule requires.

## 8. Useful commands

```
cd C:\Users\buruf\Documents\Pedmas
npm run dev          # local server on http://localhost:3080
npm test             # full test suite (139 tests)
npm run build        # production build check (stop the dev server first)
npx tsx scripts/preview-weekly-email.ts # preview the weekly email locally
```

This manual describes the platform as of August 20, 2026 (commit 9541f50). When behaviour changes, regenerate or amend it — a manual that has drifted from the product is worse than none.











